

Review Questions

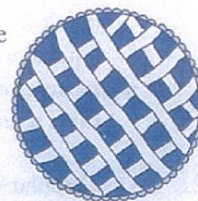
CHAPTER 3: HEATING AND COOLING

3.1 Heat

- Distinguish between the terms:
(a) heat (b) internal energy (c) temperature.
- During a physics experiment David was heard to ask Melissa, "How much heat is possessed by that piece of lead?" Melissa replied, "It is not possible to determine the amount of heat possessed by the piece of lead."
Explain why Melissa answered David in this way.
- A gas oven is said to have sufficient heat at 170°C in which to place a batch of cakes. Is the term 'heat' used correctly here? Explain.
- Ric placed two identical pieces of lead, both at room temperature, one at a time and three minutes apart into a single beaker of water also at room temperature. Ric thought that by doubling the amount of lead and therefore doubling the amount of internal energy possessed by the two pieces combined, the water temperature would change. Does the water temperature change?
- If a material has a property that changes with temperature then it may be able to be used as a thermometer. List three different properties of materials which change with temperature.
- Two types of thermometer are (a) a mercury-in-glass thermometer and (b) a platinum resistance thermometer. What property of each changes to enable it to measure temperature?
- (a) List the fixed points for the (i) Celsius scale and (ii) Kelvin scale.
(b) What do the fixed points for the Celsius scale represent?
- Convert the following temperatures.
(a) to Kelvin: (i) 37°C (ii) -20°C (iii) 114°C
(b) to Celsius: (i) 180 K (ii) 293 K (iii) 350 K
- Julie was carrying out an experiment using melting ice to determine the location of the ice point on a mercury-in-glass column. She took the precaution of using ice made of pure water. Why was it necessary for her to take this precaution?

2.2 Heat Capacity and Latent Heat

- Cathy cooked a jam tart to have for dessert. Soon after removing the tart from the hot oven she ate a piece and was surprised to find that the jam appeared to be much hotter than the surrounding pastry and burnt the inside of her mouth.
(a) Was the jam at a higher temperature than the pastry when the tart was removed from the oven? Why?
(b) Explain why the jam burnt Cathy's mouth.



11. During a physics experiment, Andrew heated equal masses of water and paraffin oil which were contained in separate but identical containers. Both were heated by the same Bunsen burner for a period of 4 minutes. Predict which liquid Andrew would find to have the highest temperature after 4 minutes. Why?
12. A constant volume of hydrogen gas has a specific heat capacity of $14300 \text{ J kg}^{-1} \text{ K}^{-1}$. Hydrogen gas is sometimes used in enclosed electrical generators. To what use do you think the hydrogen gas is being put in the generator?
13. 1.50 kg of water at 25.0°C is heated in an electric kettle for 2.00 minutes. If in this time the kettle supplies $3.00 \times 10^5 \text{ J}$ to the water calculate the temperature of the water after the 2.00 minutes interval.
14. A hot water system tank contains 150 L of water at a temperature of 20°C .
 - (a) Determine the energy required to heat the water to 65.0°C ($1 \text{ L water} = 1 \text{ kg water}$).
 - (b) If the heat is being supplied at a rate of $7.00 \times 10^3 \text{ J per second}$, how long will it take to heat the water to 65°C ?
15. Mark placed 50.0 g of iron with a temperature of 90.0°C into an insulated non-conducting container with 30.0 mL of water at 20.0°C . He then measured the new temperature of the water while doing his best to prevent heat entering and leaving the container. Calculate the final temperature of the mixture.
($c_{\text{iron}} = 4.60 \times 10^2 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$)
16. If Mark repeats the experiment but uses an insulated 200 g aluminium calorimeter cup, also at 20.0°C , calculate the final temperature of the mixture.
($c_{\text{aluminium}} = 9.00 \times 10^2 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$)
17. As a solid melts, the latent heat of fusion does not increase the kinetic energy (temperature) of the molecules. How is this heat supplied to the solid used?
18. A 3.00 kg metal electric kettle contains $1.00 \times 10^3 \text{ mL}$ of water and both are at room temperature (20.0°C). The kettle is switched on and left on for 10.0 minutes. How much water is left in the kettle if heat is supplied at $2.00 \times 10^3 \text{ J s}^{-1}$. Ignore heat loss to the surroundings.
 $(c_{\text{kettle}} = 4.00 \times 10^2 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1})$
 $c_{\text{water}} = 4.18 \times 10^3 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$,
 $L_{\text{steam}} = 2.26 \times 10^6 \text{ J kg}^{-1} \text{ }^\circ\text{C}^{-1}$)
19. During the half-time interval at a game of hockey which was played on a warm day, Kerry found that although her arms and legs were sweaty, the gentle breeze made her feel cool. Is it actually the breeze alone which is cooling her body? Explain your answer.
20. Kelvin had a soft drink at room temperature. To cool the drink he could either add 50 g of cold water at 0°C or an equivalent mass of ice pieces also at 0°C . Which would be more effective in cooling the drink - the water or the ice? Give a reason for your answer.
21. Calculate the heat required to melt 50.0 g of ice at 0°C to water at 0°C . ($L_{\text{ice}} = 3.36 \times 10^5 \text{ J kg}^{-1}$)



22. A plastic insulated drinking vessel contains 300 mL of water at 20.0°C . If 50.0 g of ice at 0°C is added to the water, calculate the final temperature of the water. (Ignore any heat transfer between the mixture and the vessel.)

23. Steam at 100°C can cause a more severe burn to your arm than an equal mass of water at the same temperature. Explain why this is the case.

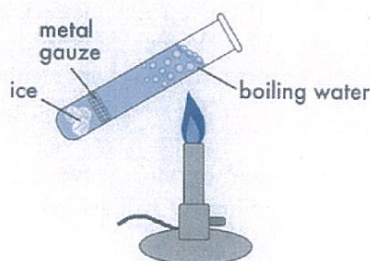


2.3 Heat Transfer

24. Liquid sodium is used in some nuclear reactors as a coolant. What physical property of sodium enables sodium to be used in this way?

25. Explain how heat is conducted in gases, liquids and metals.

26. When Adrian applied the Bunsen burner as shown in the diagram he noticed that the water boiled while the ice, held down by gauze metal, only melted very slowly. How would you explain to Adrian why this occurs?



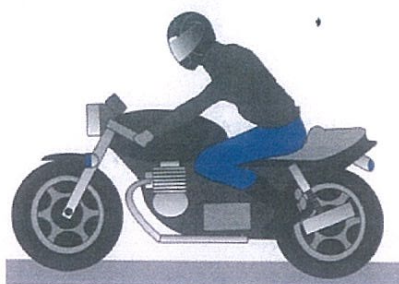
27. When the weather is cold birds can be seen fluffing up their feathers which can make them appear larger than they really are. Explain how doing this to its feathers can keep a bird warm.

28. A woollen coat does not supply heat to our body but can keep us warm on a cold day. Explain why the coat is able to keep us warm.

29. Modern brick homes have insulating material between the inside and outside walls. This has the effect of keeping the homes warm during winter. Explain why the homes keep warm.

30. Heat can be transferred through a liquid by both conduction and convection. How are the two processes different?

31. Motor bike engines have been designed with cooling fins to prevent them from overheating. In some models the fins are made vertical. Explain why they are made vertical.



32. Briefly describe how a vacuum flask, containing hot water, is designed to reduce heat loss on a cold day through the processes of conduction, convection and radiation.

33. When standing in front of a fire in an open fire place, which process of heat transfer is largely responsible for the warmth you receive?

34. Joan and Peta were walking to school one winter's morning. Joan was wearing woollen slacks and a woollen jumper but Peta, who had a phys. ed. class period one, was wearing a sports skirt and short sleeved top. A cold wind started to blow causing Peta to shiver but not Joan. Explain why the cold wind caused Peta to shiver but not Joan.

35. Many modern kitchens have a built-in alcove in which the refrigerator is placed. Explain why refrigerator manufacturers advise that if a refrigerator is placed in an alcove then an air gap of about 10 cm should be allowed for around the sides and top of the refrigerator.

2.4 Heat and the Environment

36. The intensity of solar radiation when it reaches the Earth's upper atmosphere is about 1360 Wm^{-2} . However the intensity at ground level for a city such as Perth (latitude 32°) is significantly less, typically 900 Wm^{-2} during the middle of a clear summer day.

Explain clearly the two main reasons for this reduced intensity of solar radiation.

37. The greenhouse effect of our atmosphere is very important in keeping our Earth warmer than it would otherwise be.
- (a) Explain what is meant by the greenhouse effect and how it is caused.
 - (b) Which gases in the atmosphere contribute to this effect?
 - (c) The level of greenhouse gases in our atmosphere appears to be increasing. Of what significance is this?
38. Coal burning power stations contribute to both the greenhouse effect and thermal pollution. Briefly describe how both of these affect the environment.



39. Incandescent light globes have been shown to be only about 3% efficient in converting electrical energy into light energy.

How much light energy per second would be produced by a 60 W incandescent light globe?

Explain what happens to the remainder of the electrical energy used.

Fluorescent light globes are rated as having 15% efficiency. Determine the wattage rating for a fluorescent globe producing the same amount of light as the 60 W incandescent globe.

40. In Europe, valleys between the very high snow covered mountains can receive very cold convection currents from the mountain slopes. Suggest how these convection currents may be formed.

(b) On entering the water your body (at 37°C) transfers heat to the cooler ocean water and feels cold. Even though the ocean has in total more internal energy the average kinetic energy of the water particles is less than those of your body. Hence heat is continually transferred from you to the ocean water.

3.4

(a) The hand from the icy water would feel cold whereas that from the warm water would feel warm.

(b) The skin of the hand in the icy water would have reached a temperature close to zero while held there. When placed in the tap water which is at a comparatively higher temperature heat would flow to the hand giving the sensation of warmth. The reverse occurs to the hand going from warm to cold.

3.5

The fans keep air moving past your skin. This has a cooling effect as it aids the evaporation of moisture from your body by continually removing air saturated with water vapour from your sweat.

3.6

- Wet clothes will dry out even on a cold day.
- Water spilt on the floor will eventually disappear (evaporate), even on cold days.

3.7

Wind increases the evaporation of water off a wet body. This evaporation draws a lot of heat from the body (latent heat of vaporisation) and will eventually lead to hypothermia.

3.8

Evaporative air conditioners depend on the evaporation of water to create a cooling effect. On humid days evaporation is more difficult to achieve and hence they are not as effective.

3.9.

The room will not change in temperature. The refrigerator is simply a heat pump, drawing heat from inside the fridge and letting it out behind the fridge. If the fridge door is open there is no net effect. In fact the temperature of the room may increase because the energy used by the motor and pump will be converted to heat energy.

3.10

(a) Heat from the fire reaches your face and hands by radiation. Other parts of your body not facing the fire are shielded from radiation and feel cool.

(b) Hot air near and above the burning logs is less dense than cold air and rises. As it rises it causes a vertical flickering of the flames.

(c) Wood is a very poor conductor of heat. One end of

the stick can be quite cold while the other end is actually burning.

3.11

Air is an excellent insulator. Also trapped air is not able to move and transfer heat by convection.

3.12

(a) The earth is at a much lower temperature than the sun. Cooler bodies radiate longer wavelengths.

(b) The earth's temperature would be much lower since much of the heat that the atmosphere now re-radiates back to earth would escape into space. In effect the greenhouse effect of the atmosphere would be reduced.

3.13

In the late afternoon the sun's rays reach us more obliquely. This means that they pass through more atmosphere, hence more absorption, and they cover a larger surface area in the ground, hence less intensity.

3. Review Questions

1. (a) Heat is a form of energy which is transferred from one place to another due to a difference in temperature between them.

(b) Internal energy refers to the sum of the kinetic and potential energies of the molecules in an object.

(c) The temperature of an object is its degree of hotness (or coldness) on a chosen temperature scale. Temperature is a measure of the average kinetic energy of the individual molecules in the object.

2. The piece of lead possesses internal energy not heat. Heat is energy transferred from the lead as a result of change in temperature (eg. a piece of lead heated by a Bunsen burner then placed in a beaker of cold water - heat is transferred from the lead to the water because of their difference in temperature).

3. No. Correctly speaking, the term 'internal energy' should have been used.

4. No. The internal energies of the lead and water may be different but both the water and the lead are at the same temperature. Heat will only be transferred from one to the other if they each have a different temperature.

5. Electrical resistance, voltage generated at junction of two metals, expansion.

6. (a) The mercury-in-glass thermometer uses the property that the length of a column of mercury changes with a change in temperature.

(b) The platinum resistance thermometer uses the property that the electrical resistance of platinum changes with a change in temperature.

7.

- (a) (i) ice point, 0°C ; steam point, 100°C
(ii) absolute zero, 0 K ; ice point, 273 K ,
steam point, 373 K

(b) ice point – the temperature of melting pure ice
steam point – the temperature at which pure water can exist in equilibrium with its vapour (at standard atmospheric pressure)

8.

- (a) (i) 310 K (ii) 253 K (iii) 387 K
(b) (i) -93°C (ii) 20°C (iii) 77°C

9. The ice must be pure. Impurities will lower the freezing point of water.

10.

- (a) They were both at the same temperature.
(b) The jam has a higher specific heat capacity than the pastry and can transfer a larger amount of heat to the tongue, compared to pastry, when it is tasted. (It is the water content of the jam which is responsible for the higher specific heat – and burning Cathy's tongue!)

11. Paraffin oil, as it is most likely to have a smaller specific heat capacity than water.

12. As a coolant due to its very high specific heat capacity.

13. $m = 1.50\text{ kg}$
 $T_1 = 25.0^{\circ}\text{C}$
 $Q = 3.00 \times 10^5\text{ J}$
 $c = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$
 $T_2 = ?$

$$Q = mc\Delta T$$

$$3.00 \times 10^5 = 1.50 \times 4.18 \times 10^3 \times \Delta T$$

$$\Delta T = \frac{3.00 \times 10^5}{1.50 \times 4.18 \times 10^3}$$

$$\therefore \Delta T = 47.8^{\circ}\text{C}$$

$$\therefore \text{Temperature after 2 mins} = 47.8 + 25.0$$

$$= 72.8^{\circ}\text{C}$$

14. $m = 150\text{ kg}$
 $T_1 = 20.0^{\circ}\text{C}$
 $T_2 = 65.0^{\circ}\text{C}$
 $c = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$
 $Q = ?$

(a) $Q = mc\Delta T$
 $= 150 \times 4.18 \times 10^3 \times 45.0$
 $= 28.2 \times 10^6\text{ J}$
 $(= 28.2\text{ MJ})$

(b) $\text{Time} = \frac{2.82 \times 10^7}{7.00 \times 10^3}$
 $= 4031\text{ s}$
 $= 67.2\text{ minutes}$

15. $T_{1\text{ iron}} = 90.0^{\circ}\text{C}$
 $T_{2\text{ iron}} = T$
 $T_{1\text{ water}} = 20.0^{\circ}\text{C}$
 $T_{2\text{ water}} = T$
 $m_{\text{water}} = 30.0\text{ g} = 0.030\text{ kg}$
 $m_{\text{iron}} = 50.0\text{ g} = 0.050\text{ kg}$
 $c_{\text{iron}} = 4.60 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$
 $c_{\text{water}} = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$

Heat lost by iron = Heat gained by water

$$mc\Delta T_{\text{iron}} = mc\Delta T_{\text{water}}$$

$$0.050 \times 4.60 \times 10^2 \times (90.0 - T) =$$

$$0.030 \times 4.18 \times 10^3 \times (T - 20.0)$$

$$2070 - 23T = 125.4T - 2508$$

$$148.4T = 4578$$

$$T = 30.8^{\circ}\text{C}$$

$$\therefore \text{Final temperature} = 30.8^{\circ}\text{C}$$

16.

In addition to No. 15

$$T_{1\text{ alum}} = 20.0^{\circ}\text{C}$$

$$T_{2\text{ alum}} = T$$

$$m_{\text{alum}} = 0.200\text{ kg}$$

$$c_{\text{alum}} = 9.00 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$$

Heat lost by iron = Heat gained by water + heat gained by aluminium

$$mc\Delta T_{\text{iron}} = mc\Delta T_{\text{water}} + mc\Delta T_{\text{alum}}$$

$$0.050 \times 4.60 \times 10^2 \times (90.0 - T) =$$

$$0.030 \times 4.18 \times 10^3 \times (T - 20.0) +$$

$$0.200 \times 9.00 \times 10^2 \times (T - 20.0)$$

$$2070 - 23T = 125.4T - 2508 + 180T$$

$$-3600$$

$$328.4T = 8178$$

$$T = 24.9^{\circ}\text{C}$$

$$\therefore \text{Final temperature} = 24.9^{\circ}\text{C}$$

17. The heat is used to overcome attractive forces between the molecules in the solid. This loosens the molecules from their 'fixed' positions to allow them to 'flow' over each other.

18. $m_{\text{kettle}} = 3.00\text{ kg}$
 $c_{\text{kettle}} = 4.00 \times 10^2\text{ J kg}^{-1}\text{C}^{-1}$
 $\Delta T_{\text{kettle}} = 80.0^{\circ}\text{C}$
 $m_{\text{water}} = 1.00\text{ kg}$
 $c_{\text{water}} = 4.18 \times 10^3\text{ J kg}^{-1}\text{C}^{-1}$
 $\Delta T_{\text{water}} = 80.0^{\circ}\text{C}$
 $L_v = 2.26 \times 10^6\text{ J kg}^{-1}$
 $m_s = ?$
 $Q = 2.00 \times 10^3\text{ J s}^{-1}$
Ht supplied = Ht of kettle + Ht of water

$$\begin{aligned}
 &+ H_f \text{ of Vap'n} \\
 Q &= m c \Delta T_k + m c \Delta T_w + m L_v \\
 10.0 \times 60.0 \times 2.00 \times 10^3 &= \\
 (3.00 \times 400 \times 80.0) + (1 \times 4180 \times 80.0) + & \\
 m \times 2.26 \times 10^6 &
 \end{aligned}$$

$$m = \frac{0.770 \times 10^6}{2.26 \times 10^6}$$

$$= 0.34 \text{ kg}$$

Mass of water vaporised = $3.40 \times 10^{-1} \text{ kg}$

Mass of water left in kettle = $6.60 \times 10^{-1} \text{ kg}$

19. No. The breeze does assist in the rate at which her sweat evaporates from her body. As the sweat evaporates, the latent heat of vaporisation required is supplied by Kerry's body. This loss of heat results in the cooling sensation that she experiences.

20. Ice. As the ice melts it requires a large amount of heat (latent heat of fusion) which is additional to the heat gained by the water at 0°C in the mixture. As a consequence the soft drink will cool more effectively with the ice as most of the heat supplied to the ice will come from the soft drink.

$$\begin{aligned}
 21. \quad m &= 0.050 \text{ kg} \\
 L_f &= 3.36 \times 10^5 \text{ J kg}^{-1} \\
 Q &= ? \\
 Q &= mL \\
 &= 0.050 \times 3.36 \times 10^5 \\
 &= 1.68 \times 10^4 \text{ J}
 \end{aligned}$$

$$\begin{aligned}
 22. \quad m_{\text{ice}} &= 0.050 \text{ kg} \\
 L_f &= 3.36 \times 10^5 \text{ J kg}^{-1} \\
 c_w &= 4.18 \times 10^3 \text{ J kg}^{-1} \text{ } ^\circ\text{C}^{-1} \\
 m_w &= 0.300 \text{ kg} \\
 T_1 &= 20.0^\circ\text{C} \\
 T_2 &= T \\
 \text{Heat gained by ice} &= \text{Heat lost by water} \\
 m L_f + m c \Delta T &= m c \Delta T \\
 &= (0.050)(3.36 \times 10^5) + (0.050) \\
 &\quad (4.18 \times 10^3)(T - 0) \\
 &= (0.300)(4.18 \times 10^3)(20.0 - T) \\
 1.68 \times 10^4 + 209T &= 25080 - 1254T \\
 1463 T &= 8280 \\
 T &= 5.7^\circ\text{C} \\
 \therefore \text{Final temperature of water} &= 5.7^\circ\text{C}
 \end{aligned}$$

23. Steam at 100°C condensing to water at 100°C on the arm releases the latent heat of vaporisation acquired during the formation of the steam. It is this heat which is responsible for producing the more severe burn.

24. Liquid sodium is used as a coolant because it is a good conductor and very effective at removing heat from the reactor.

25. Gases: Collisions between gas molecules at different temperatures result in the transfer of heat from high temperature molecules to low temperature molecules. This continues until all molecules are (on average) equally energetic.

Liquids: The molecules which are at a high temperature transfer heat due to their more energetic movements by collision with cooler molecules. Heat is transferred to cooler regions of the liquid by this process.

Metals: Electrons which are free to move about the metal lattice are able to rapidly transfer energy to the cooler regions of the metal by collisions with other electrons and metal atoms.

26. Water and glass are poor conductors of heat. The transfer of heat by conduction to the ice from through the glass and water occurs slowly.

27. By fluffing its feathers the bird traps air around it beneath the outer layer of feathers. The pockets of trapped air reduce heat loss from the bird's body because air is a poor conductor of heat.

28. Pockets of air trapped between the fibres of the coat reduce heat loss from our body by conduction because both air and the fibres are poor conductors of heat.

29. Both the bricks and the insulating material have a low thermal conductivity. The insulating material slows down the rate of heat loss by conduction.

30. Conduction involves the transfer of heat as molecules of the liquid collide with cooler molecules in the liquid. Convection is the transfer of heat through the liquid by the liquid's heated molecules actually moving through the liquid. The heated portion of the liquid expands, becomes less dense than the surrounding liquid and so rises through the liquid.

31. The heat can be transferred from the fins by radiation and convection. The vertical fins can cool by convection by allowing the hot surrounding air to rise as a convection current.

32. Conduction: The flask consists of a double glass wall which contains a vacuum preventing conduction. The cork/plastic stopper is made of a material which is a poor conductor of heat.

Convection: The vacuum prevents convection. When the stopper is removed, heat can be lost by convection through the top of the flask.

Radiation: The glass walls of the flask are silvered to reduce radiation and reflect any radiated heat back into the flask.

33. Radiation. Most of the heated air rises, through convection, up the chimney.

34. Joan's woollen clothing reduced the transfer of body heat to the outside cold air, enabling her to remain warm. The low temperature around Peta's arms and legs resulted in heat loss from her body to the surroundings. Peta's body shivered to increase her metabolism so as to produce more heat to replace that which was lost to the surroundings.

35. The efficient functioning of a refrigerator relies on the heat extracted from within the refrigerator to be released outside of the refrigerator - from the coils at the back of the refrigerator. Heat dissipated from the coils must be dispersed (mainly by convection) from the coils if the refrigerator is to be efficient. An air gap around the refrigerator enables the convection to occur.

36. The intensity is less due to the effect of the atmosphere and the angle of the sun. The atmosphere always absorbs and reflects some of the solar radiation. At a latitude of 32° the radiation from the sun is spread over a larger area and this further reduces the intensity.

37. (a) The greenhouse effect is caused by gases in the atmosphere trapping heat being radiated by the Earth. This allows most of the heat that would have otherwise been lost to space to heat up the atmosphere and also be partially returned to Earth.

(b) The gases mainly responsible for the greenhouse effect are water vapour (H_2O) and carbon dioxide (CO_2). Methane (CH_4), ozone (O_3), chlorofluorocarbons (CFC's) and nitrous oxide (N_2O) also contribute.

(c) An increase in greenhouse gases would most likely contribute to global warming and cause significant changes to our environment.

38. Greenhouse effect: Radiation from the sun warms the earth's surface which radiates some of this heat back into space. However increased levels of carbon dioxide in the atmosphere (from the burning of fossil fuels eg coal) prevents some heat loss to space causing the atmospheric temperature to increase.

Thermal pollution: Water used as a coolant in the power station may be pumped as hot water into surrounding rivers and lakes causing a rise in temperature which can be detrimental to aquatic life.

39. (a) $P = 60\text{ W} = 60\text{ Js}^{-1}$

$3\% = 60 \times 3/100 = 1.8\text{ Js}^{-1}$

(b) The remainder of the energy is converted to heat. This warms the light bulb and fittings and is dissipated into the surroundings.

(c) Fluoro 5 times more efficient $(15/3) = 5$

Wattage needed $= 60/5 = 12\text{ W}$

40. Air high on the mountains gets very cold, contracts and becomes more dense. This cold air then flows down the mountain slopes into the valleys.

CHAPTER 4: ELECTRICITY

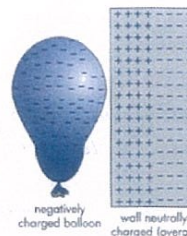
4. Chapter Questions

4.1.

Protons are located in the nucleus of atoms and are not free to move.

4.2.

The charged balloon (say negatively charged) induces an opposite charge on the surface of the wall it is near. Attraction between opposite charges results.



4.3.

$$No(e^-) = \frac{1}{1.6 \times 10^{-19}} = 6.25 \times 10^{18}$$

4.4.

$$P = \frac{W}{t} = \frac{qV}{t} = \left(\frac{q}{t}\right)(V)$$

$$\therefore P = VI$$

4.5.

As temperature rises atoms vibrate more strongly. This inhibits the flow of charge from atom to atom.

4.6.

(a) (i) At 1.5 V $I \approx 18\text{ A}$

$$\therefore R = \frac{V}{I} \approx \frac{1.5}{18} \approx 8.3 \times 10^{-2}\ \Omega$$

(ii) At 3.0 V $I \approx 23\text{ A}$

$$\therefore R = \frac{V}{I} \approx \frac{1.5}{23} \approx 6.5 \times 10^{-2}\ \Omega$$

(b)

$$P = VI = (3)(23) = 69\text{ W}$$

4.7.

(a) C would conduct most current. The greater the cross-sectional area the smaller the resistance.

(b) B would conduct least current. Resistance increases with length.

4.8.

Parallel circuits offer several paths for current to flow. Hence resistance is always lowered when resistors are placed in parallel.

4.9.

3. Review Questions

